Project 2

Mountain Explorer

Design Document

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Table of Contents

Project Description 4
Assumptions and Constraints 4
Assumptions 4
Constraints 4
Out of Scope 4
User Personas 5
Persona 1 – Dan, Experienced Hiker 5
Persona 2 – Jen, Site Administrator 5
Persona 3 – Bob, New Hiker 6
User Stories 6
User Story 1 – Dan: Create Favorites List 6
User Story 2 – Bob: Create Account and Search 6
User Story 3 – Dan: Update Favorites List 6
User Story 4 – Dan: Delete a Favorite Trail 6
User Story 5 – Bob: Update Account Password 6
User Story 6 – Jen: Delete a Trail 7
User Story 7 – Jen: Create a Trail 7
User Story 8 – Jen: Search for a Trail 7
User Story 9 – Jen: Update a Trail 7
Technology Used 8
Design Mockups 9
Mockup Page 1 – Homepage (Joseph Sikowitz) 9
Mockup Page 2 – User side - Registration (Ken Yong) 10
Mockup Page 3 – User side - Login page (Ken Yong) 11
Mockup Page 4 – User side - Update password page (Ken Yong) 12
Mockup Page 5 – User side - Profile page (Ken Yong) 13
Mockup Page 6 – User side - Search mountains page (Ken Yong) 14
Mockup Page 7 – User side - Trails display page (Ken Yong) 15
Mockup Page 8 – User side - Favorite page (Ken Yong) 16
Mockup Page 9 – Admin side – Admin Login (Joseph Sikowitz) 17
Mockup Page 10 – Admin side – Admin Add Mountain (Joseph Sikowitz) 18
Mockup Page 11 – Admin side – Admin Search to Edit or Delete Mountain (Joseph Sikowitz) 19
Mockup Page 12 – Admin side – Admin Edit Mountain (Joseph Sikowitz) 20
Mockup Page 13 – Admin side – Admin Add Trail (Joseph Sikowitz) 21
Mockup Page 14 – Admin side – Admin Search to Edit or Delete Trail (Joseph Sikowitz) 22

Mockup Page 15 – Admin side – Admin Edit Trail (Joseph Sikowitz) 23
Document Control 24

Project Description

Mountain Explorer is a web application designed to help hikers discover, organize, and manage hiking information. Users can create and manage an account, search for mountains and their associated trails, and save trails to a personalized favorites list. The favorites feature allows users to keep track of trails that they have completed as well as trails they plan to hike in the future.

This web application also provides administrative functionality for maintaining mountain and trail information. Administrators can create, view, update, search, and delete mountain and trail records so that the information available to users can be maintained and updated.

Assumptions and Constraints

This section details the boundaries of the project.

Assumptions

- Users have access to a modern web browser and an internet connection.
- Users provide valid account information when registering.
- Each user is identified by a unique email address.
- Authentication is simulated for the scope of this project rather than implemented as a production level authentication system.

Constraints

- The frontend must use HTML5, CSS3, and Vanilla JavaScript with ES6 modules.
- The backend must use Node.js and Express.
- MongoDB must be accessed using the native MongoDB Node.js driver.

Out of Scope

- Production-level user authentication and security.
- Password recovery and email verification.
- Password hashing is out of scope for this project. Passwords are stored as plain text in MongoDB because secure authentication is not required for this project.

User Personas

Personas and user stories for the three audience groups identified for the Mountain Explorer web application.

Persona 1 – Dan, Experienced Hiker

Background

Dan is an experienced hiker who has been hiking for over 20 years. He regularly explores different mountains and trails and wants a convenient way to keep track of his hiking activities.

Goal

He wants to keep track of trails he has already completed and organize trails that he would like to hike in the future.

Need

He needs an easy way to save trails to his favorites list, update the information in his list as his hiking plans change, and remove trails that he is no longer interested in.

Persona 2 – Jen, Site Administrator

Background

Jen is a site administrator who is responsible for maintaining the mountain and trail information available in the Mountain Explorer application.

Goal

She wants to keep the mountain and trail information current so that users can find useful information when planning their hikes.

Need

She needs an efficient way to create, search, view, update, and delete mountain and trail records. This allows her to add new trails, update existing information, and remove trails that are no longer available.

Persona 3 – Bob, New Hiker

Background

Bob is new to hiking and wants to learn more about hiking. He does not have the same level of experience as Dan and wants to find mountains and trails that are suitable for a beginner.

Goal

He wants to create an account, search for beginner-friendly mountains and trails, and organize the trails that interest him.

Need

He needs an easy way to manage his account, search for mountains and trails, and save trails that he may want to hike in the future.

User Stories

User Story 1 – Dan: Create Favorites List

As a hiker, Dan wants to create a list of trails that he has hiked in the favorites list so that he can remember all of the hikes he has completed.

Assigned to: Ken Yong

User Story 2 – Bob: Create Account and Search

As a new hiker, Bob wants to create an account so that he can search for easy mountains to hike in his area.

Assigned to: Ken Yong

User Story 3 – Dan: Update Favorites List

As an experienced hiker, Dan wants to update his favorites list to add trails he wants to hike so that he can plan his future trips.

Assigned to: Ken Yong

User Story 4 – Dan: Delete a Favorite Trail

As a hiker, Dan wants to delete a trail that he has decided he no longer wants to attempt so that his favorites list remains up to date.

Assigned to: Ken Yong

User Story 5 – Bob: Update Account Password

As a site user, Bob wants to update his account password so that his password is more complex.

Assigned to: Ken Yong

User Story 6 – Jen: Delete a Trail

As an administrator, Jen wants to delete a trail that has washed away so that users no longer attempt to use it.

Assigned to: Joseph Sikowitz

User Story 7 – Jen: Create a Trail

As an administrator, Jen wants to create a trail that has just been built so that users are aware of it.

Assigned to: Joseph Sikowitz

User Story 8 – Jen: Search for a Trail

As an administrator, Jen wants to search for a trail to see whether it has already been added so that she does not create a duplicate.

Assigned to: Joseph Sikowitz

User Story 9 – Jen: Update a Trail

As an administrator, Jen wants to update an existing trail so that users have access to the most current trail information.

Assigned to: Joseph Sikowitz

Technology Used

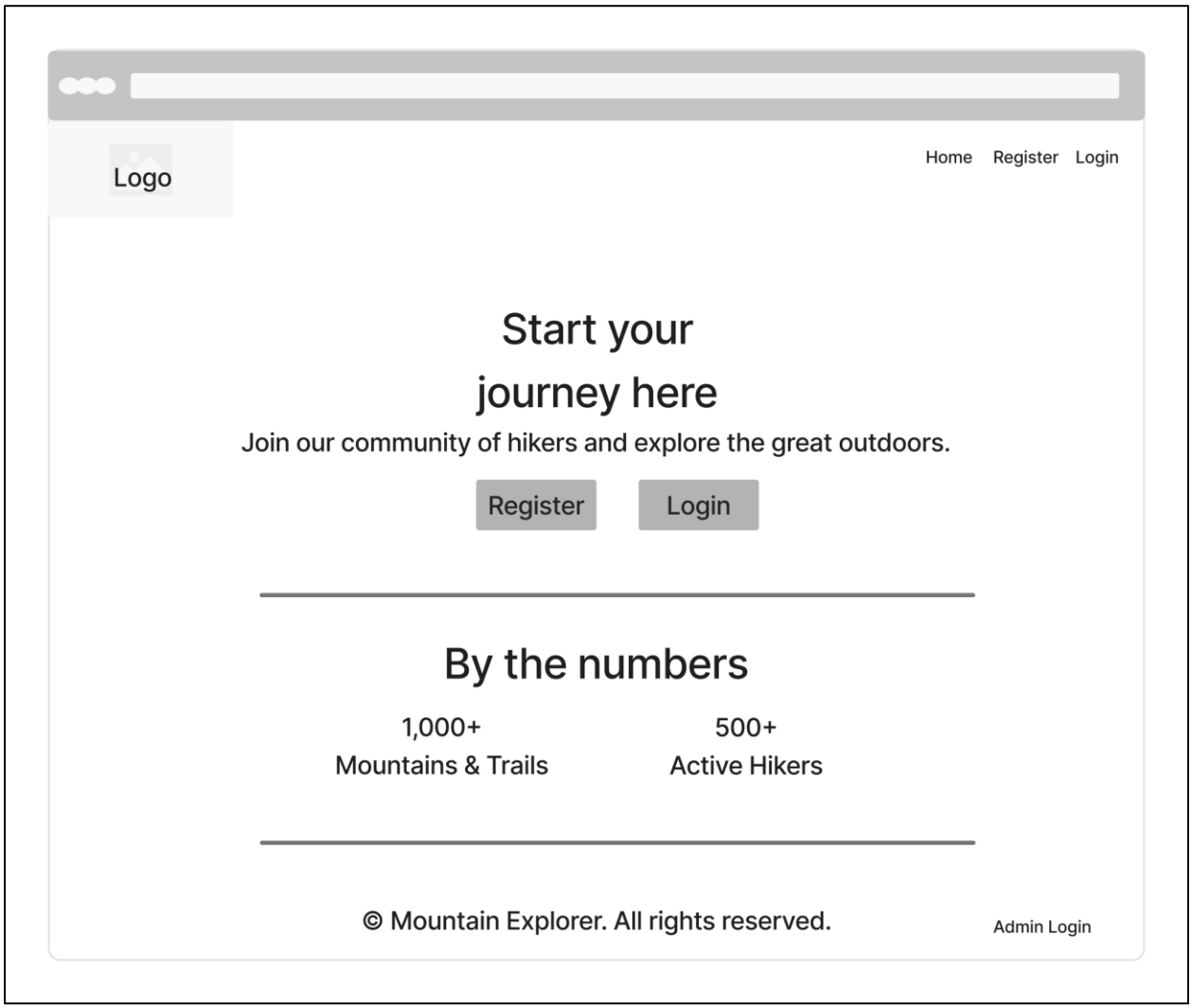
The following technologies support the design and implementation of the website.

Technology	Purpose
HTML5	Provides the semantic structure and content for the Mountain Explorer web pages, including user, favorites, mountain, trail and administrator pages.
CSS3	Provides custom visual styling and layout for the website.
Bootstrap 5	Provides responsive layout, components, and utility classes used throughout the website.
JavaScript ES6+	Provides client-side functionality, including form handling, API requests, client-side rendering, user interactions, and session management.
Node.js	Provides the JavaScript runtime environment for the server-side application.
Express	Provides backend routing and REST API endpoints used by the frontend to perform CRUD operations.
MongoDB	Stores application data, including users, favorites, mountains, and trails.
MongoDB Native Node.js Driver	Provides the connection between the Node.js backend and MongoDB and is used to query and modify database records.
dotenv	Loads environment variables used for application configuration and the MongoDB connection.
ESLint	Checks JavaScript against the class linting configuration and helps identify code-quality issues.
Prettier	Keeps project files consistently formatted.
Git & GitHub	Provides version control, supports collaboration between team members and stores the public project repository.
https://www.mockaroo.com/	Generate large amounts of realistic sample data for testing and populating the MongoDB database.
Render	Hosts and deploys the web application to make it publicly accessible online

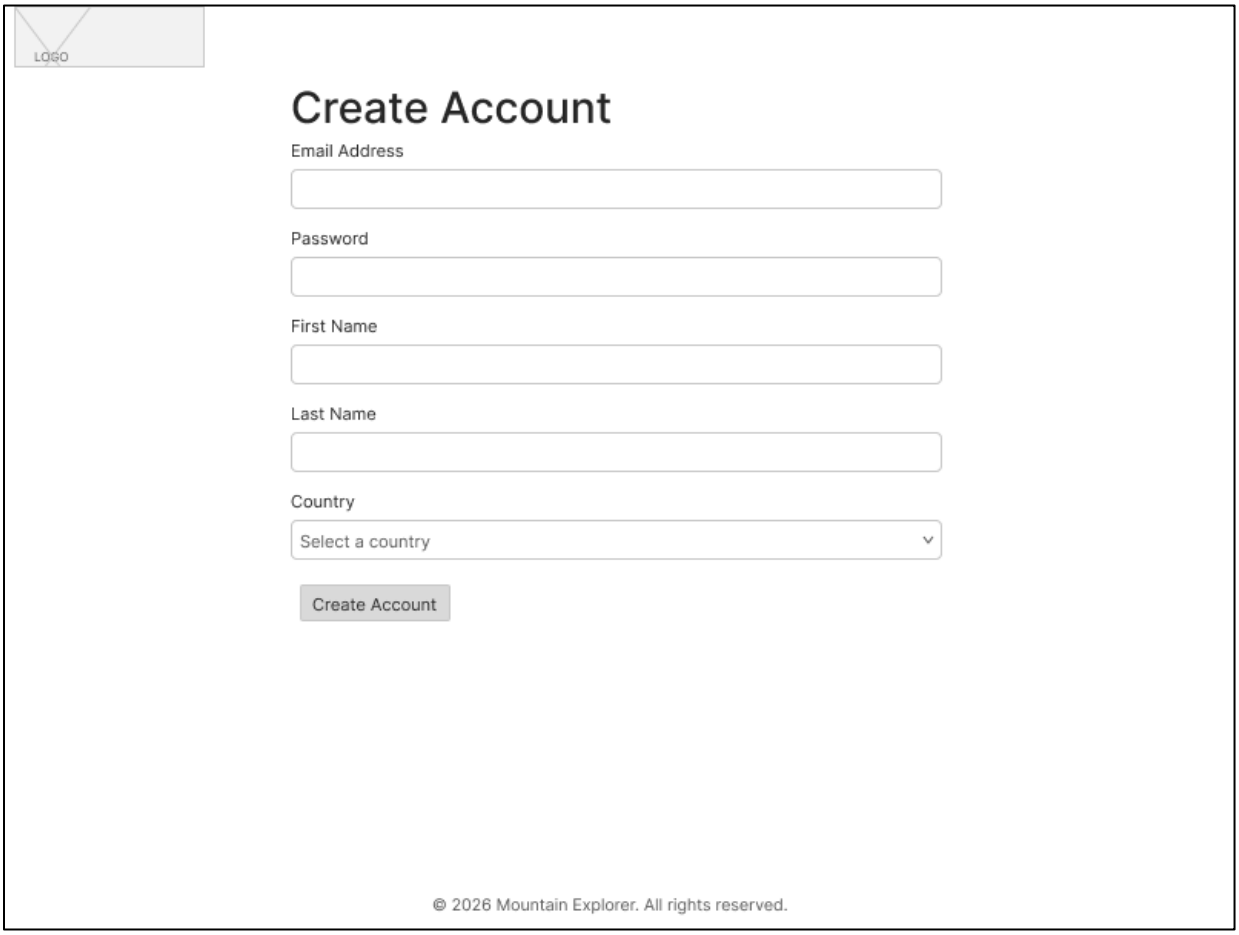
Design Mockups

These mockups were created through the Figma.com website.

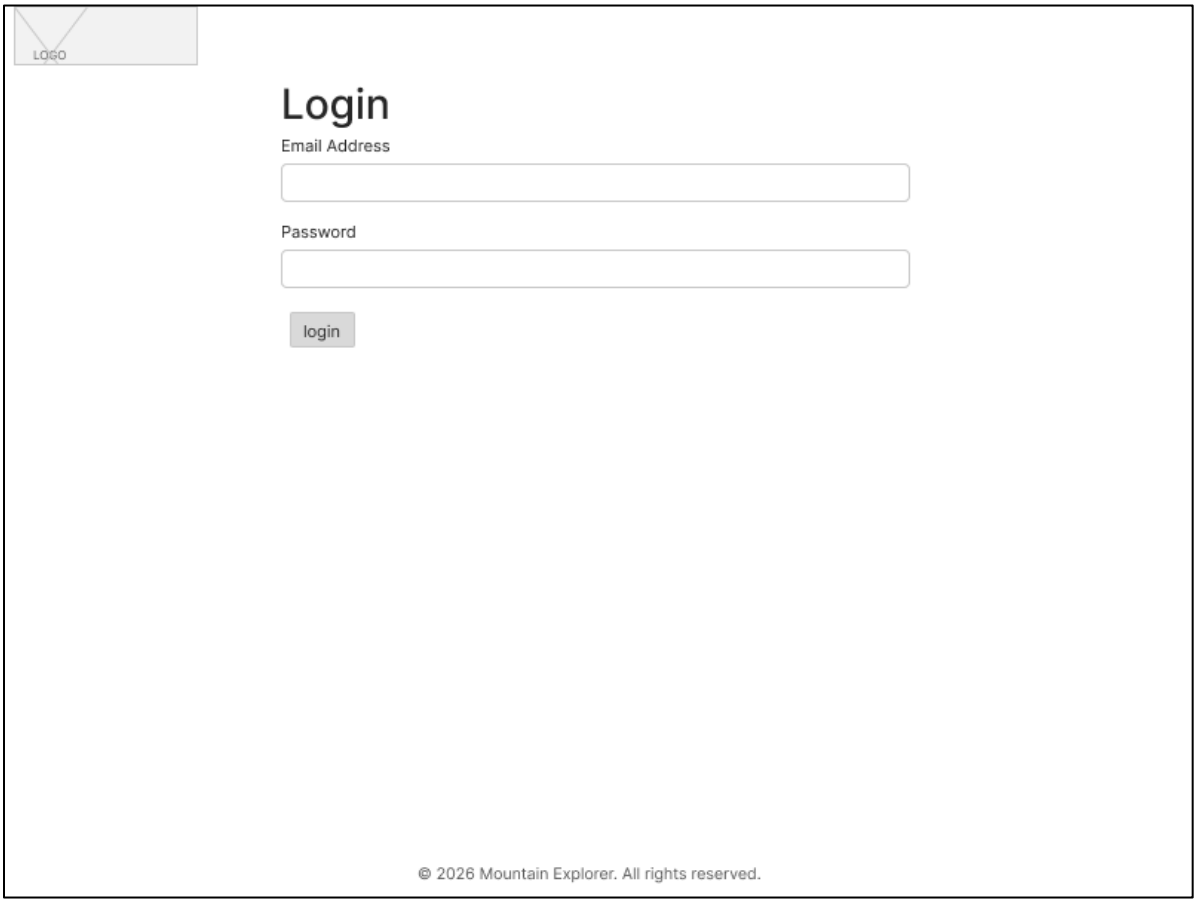
Mockup Page 1 – Homepage (Joseph Sikowitz)



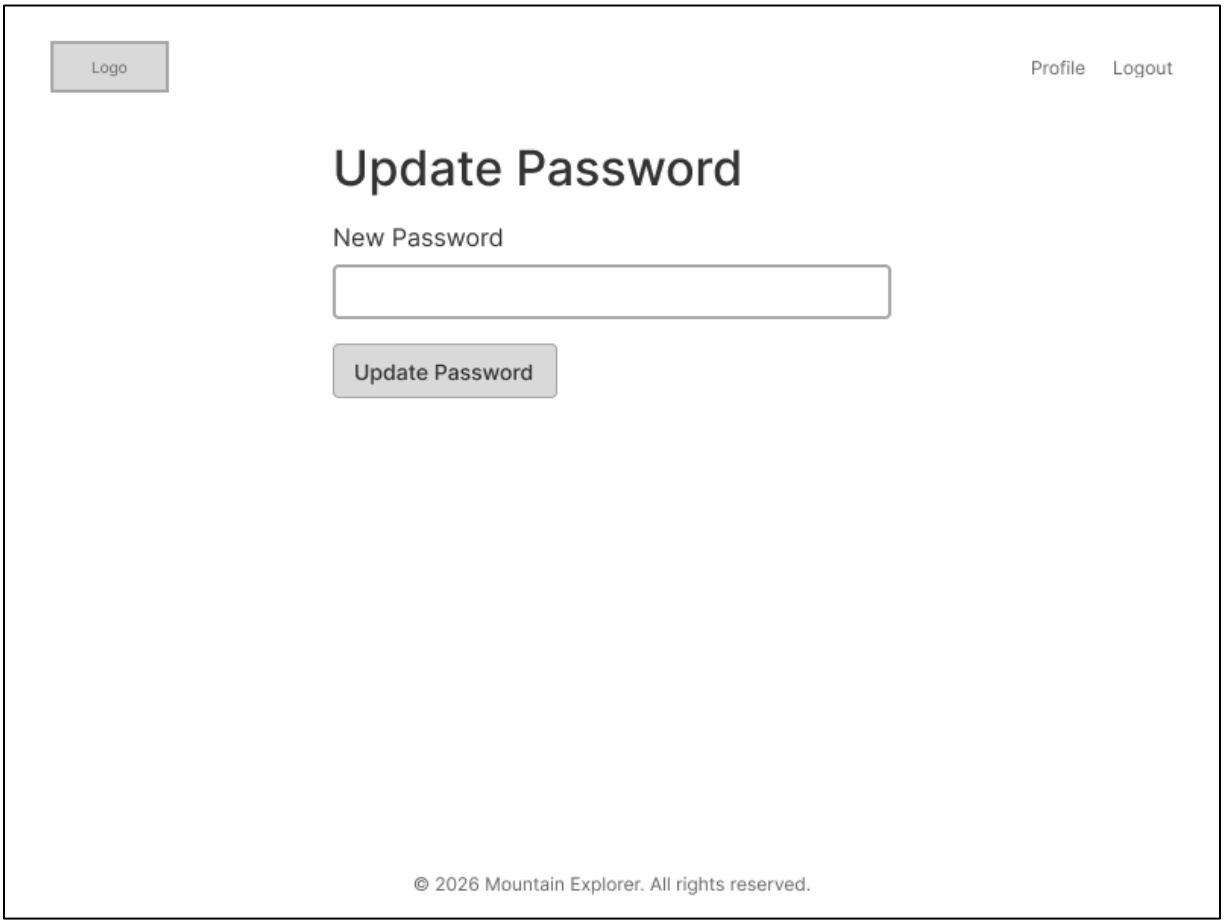
Mockup Page 2 – User side- Registration (Ken Yong)



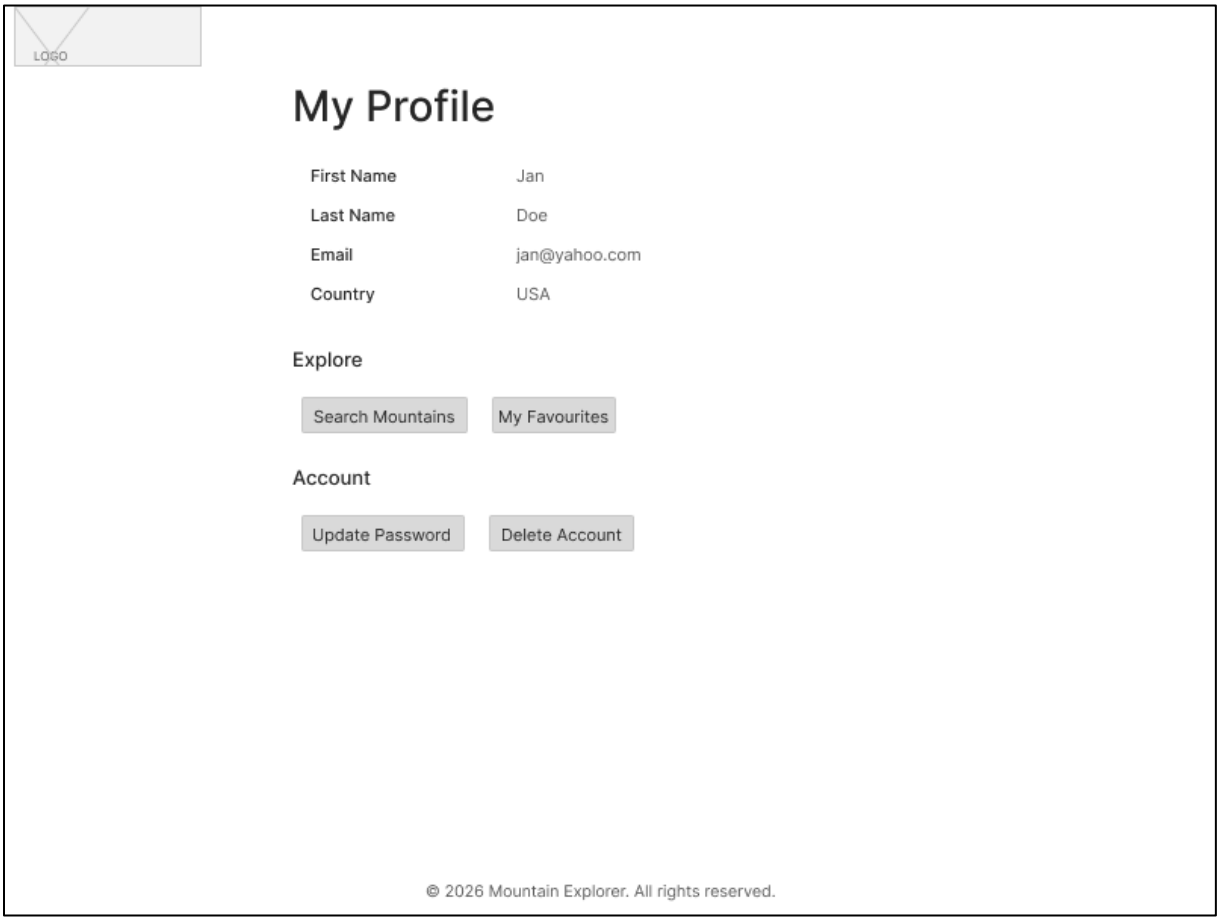
Mockup Page 3 – User side- Login page (Ken Yong)



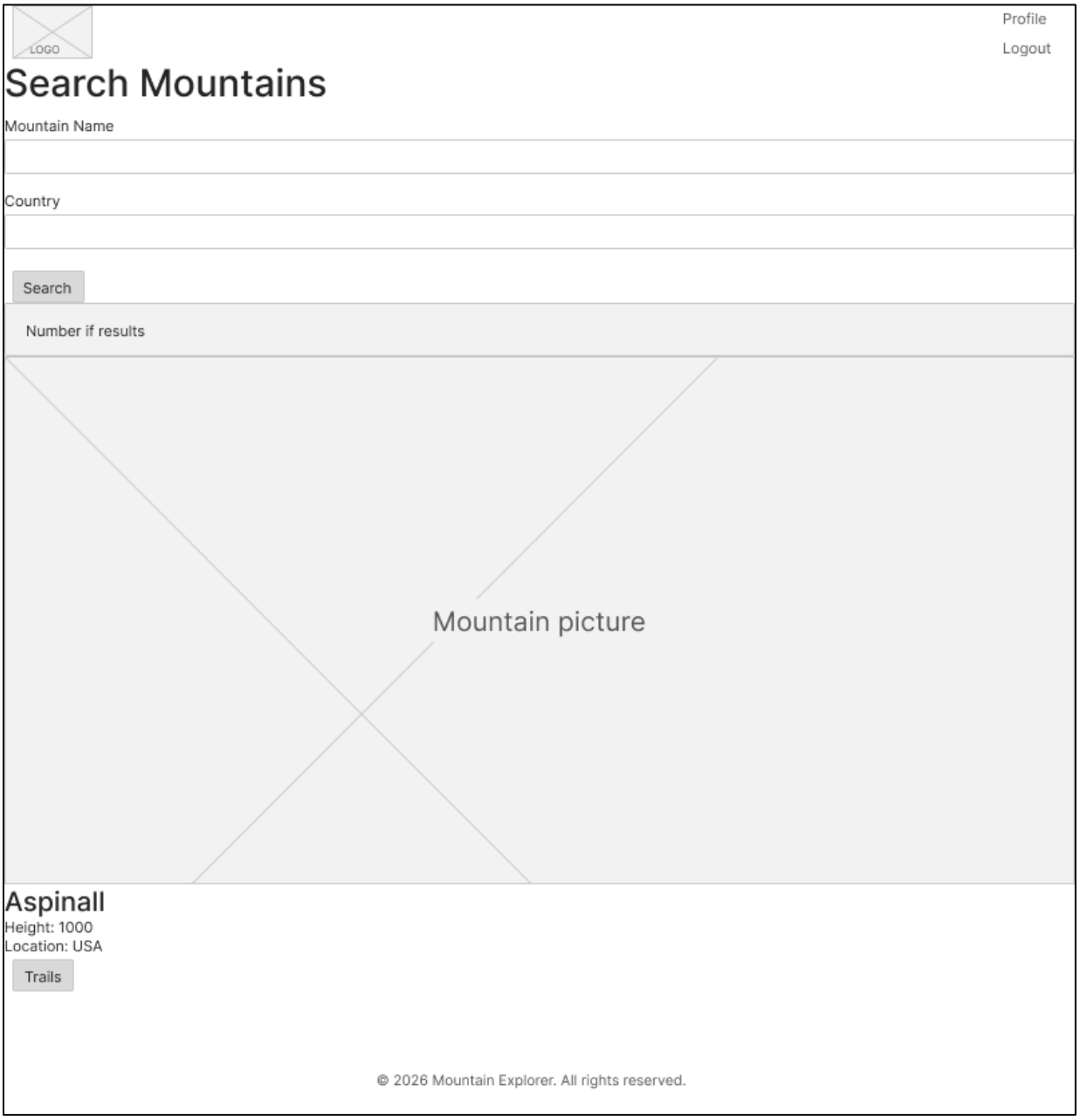
Mockup Page 4 – User side- Update password page (Ken Yong)



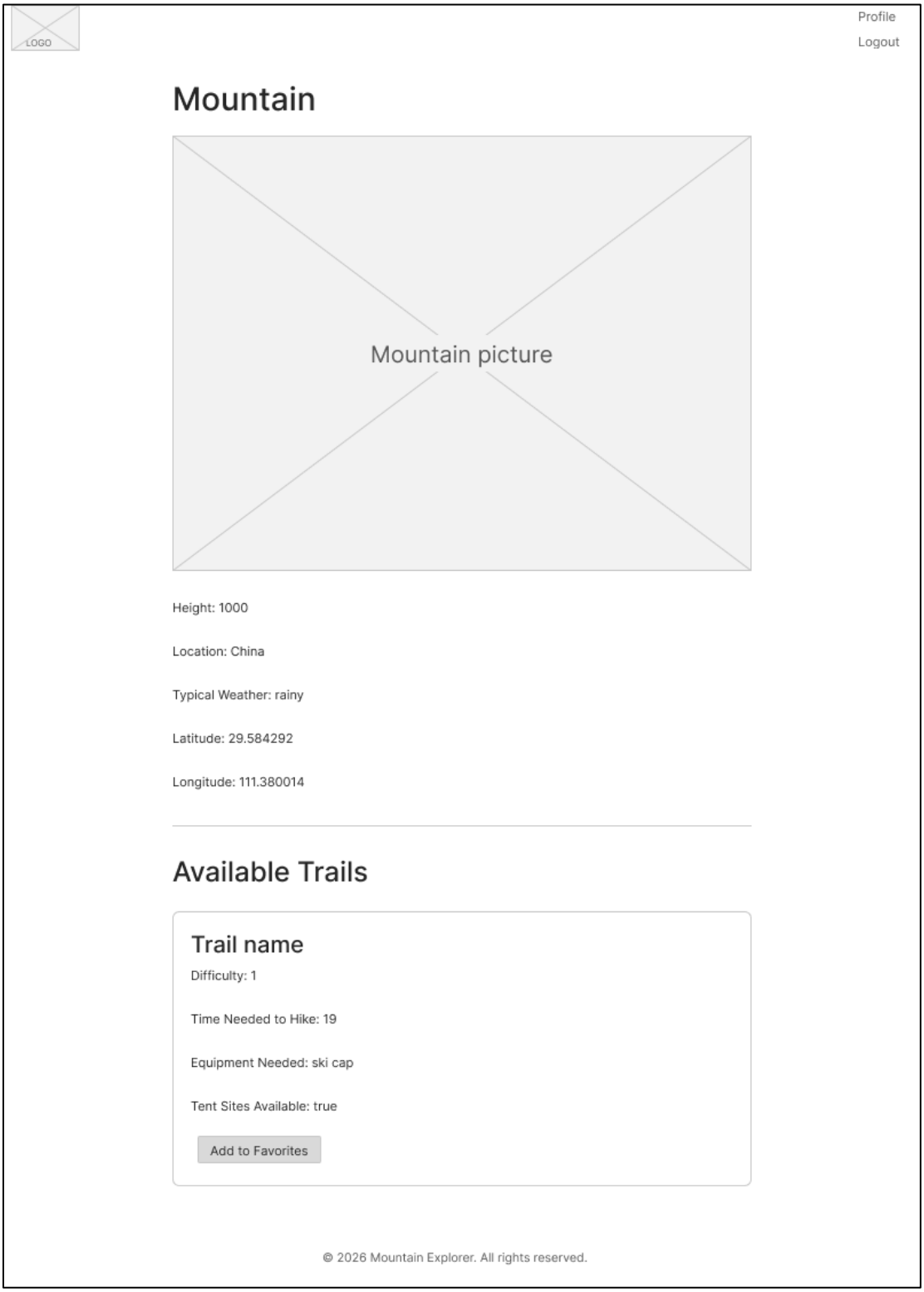
Mockup Page 5 – User side- Profile page (Ken Yong)



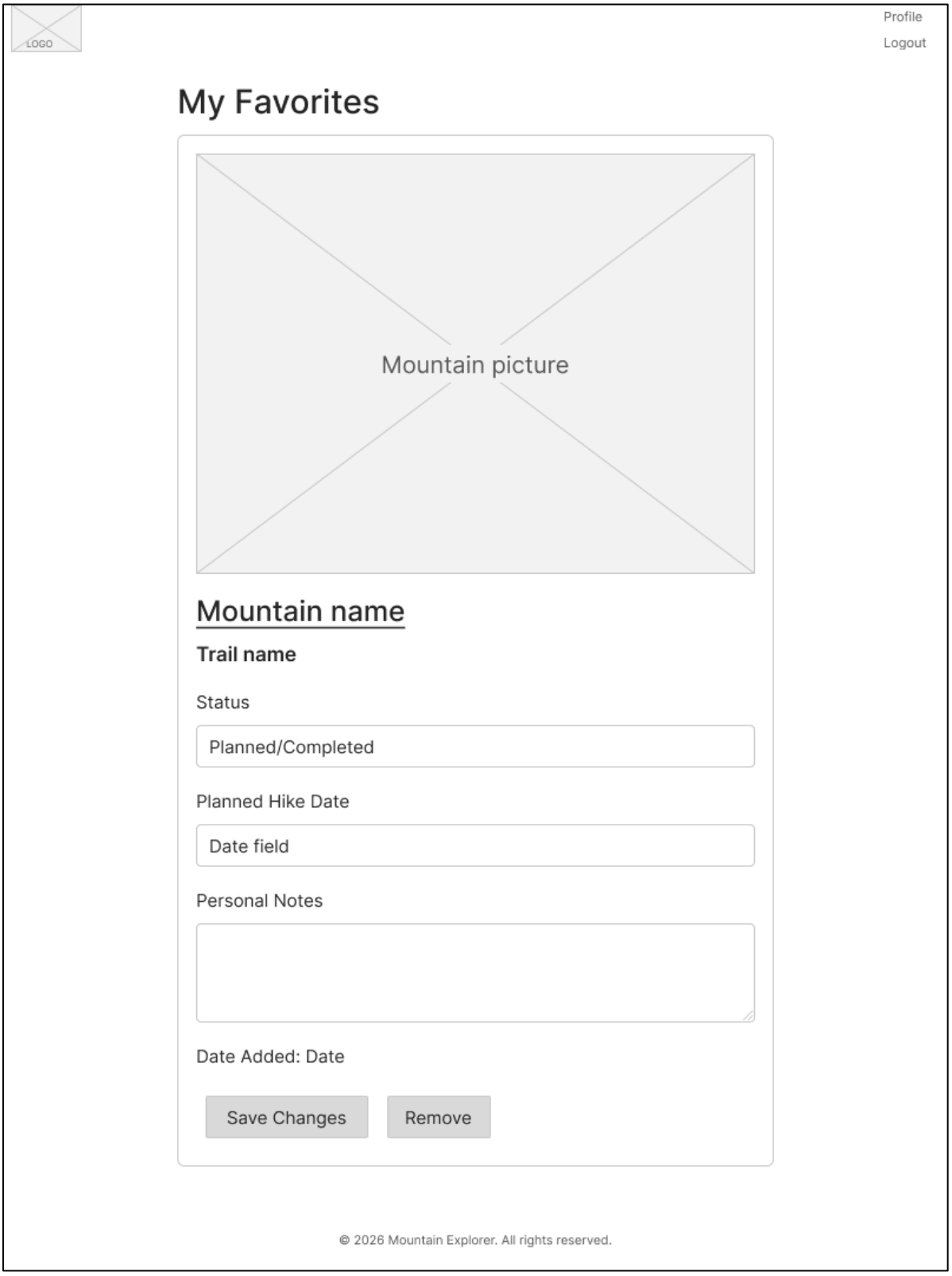
Mockup Page 6 – User side- Search mountains page (Ken Yong)



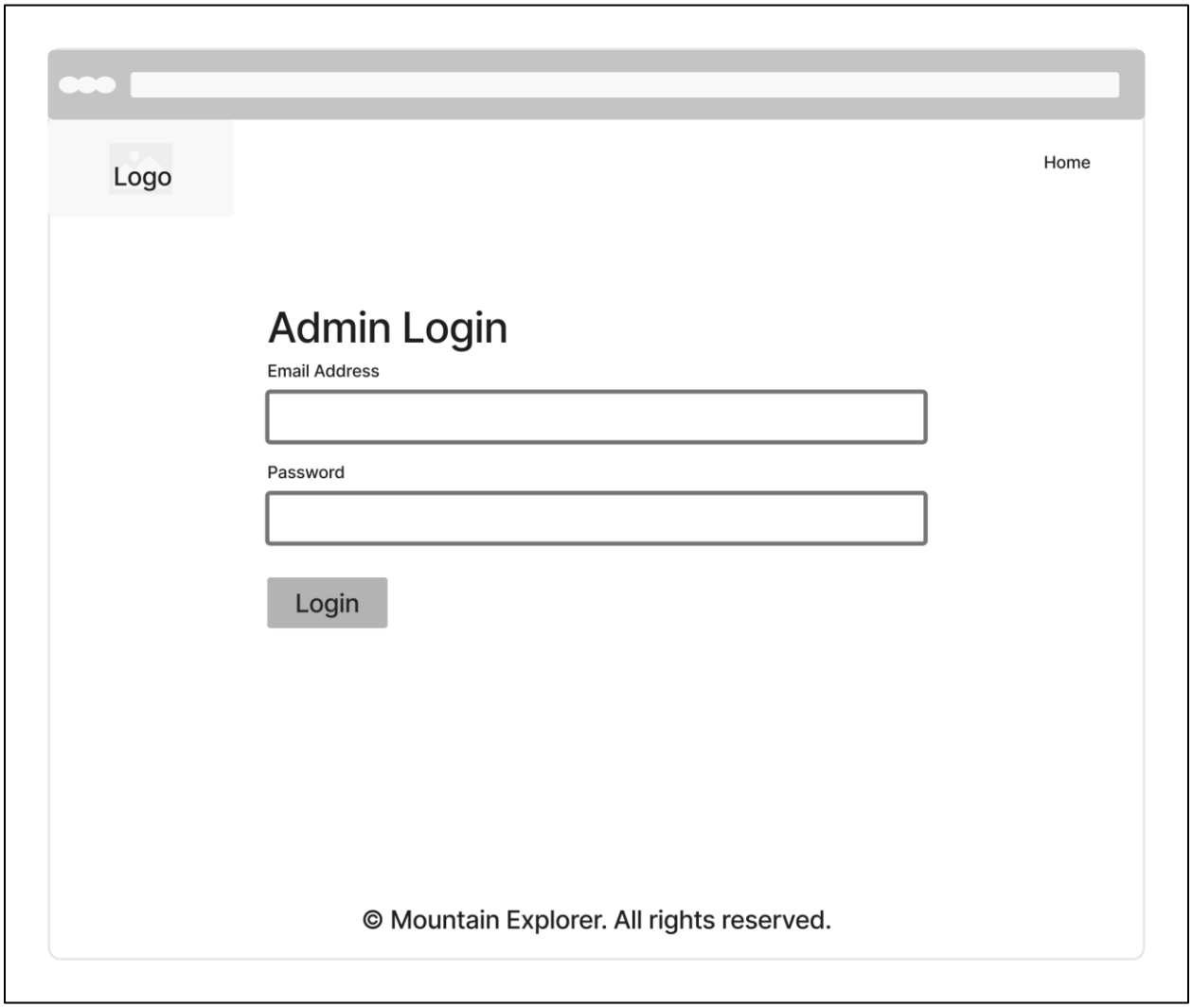
Mockup Page 7 – User side- Trails display page (Ken Yong)



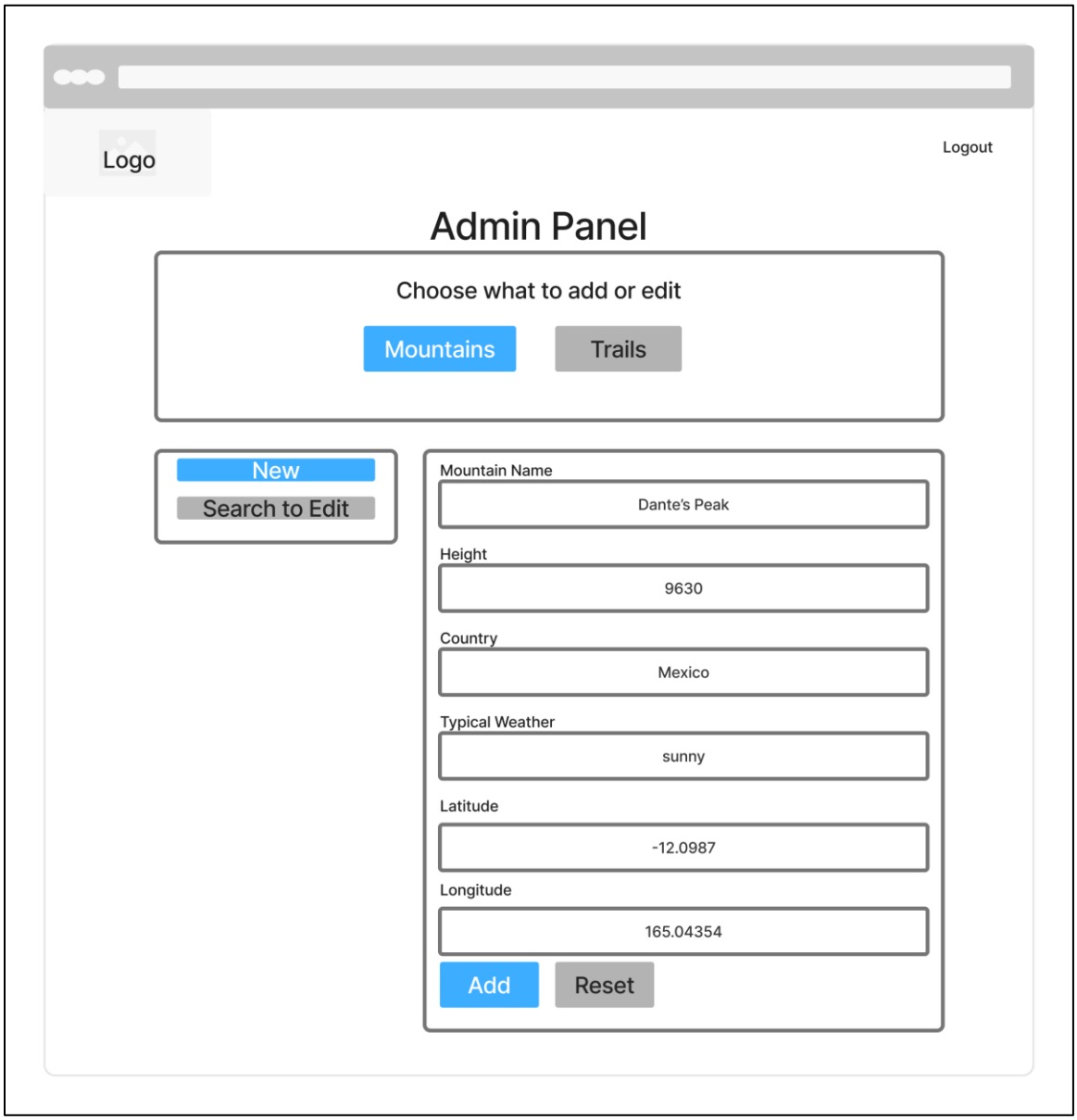
Mockup Page 8 – User side- Favorite page (Ken Yong)



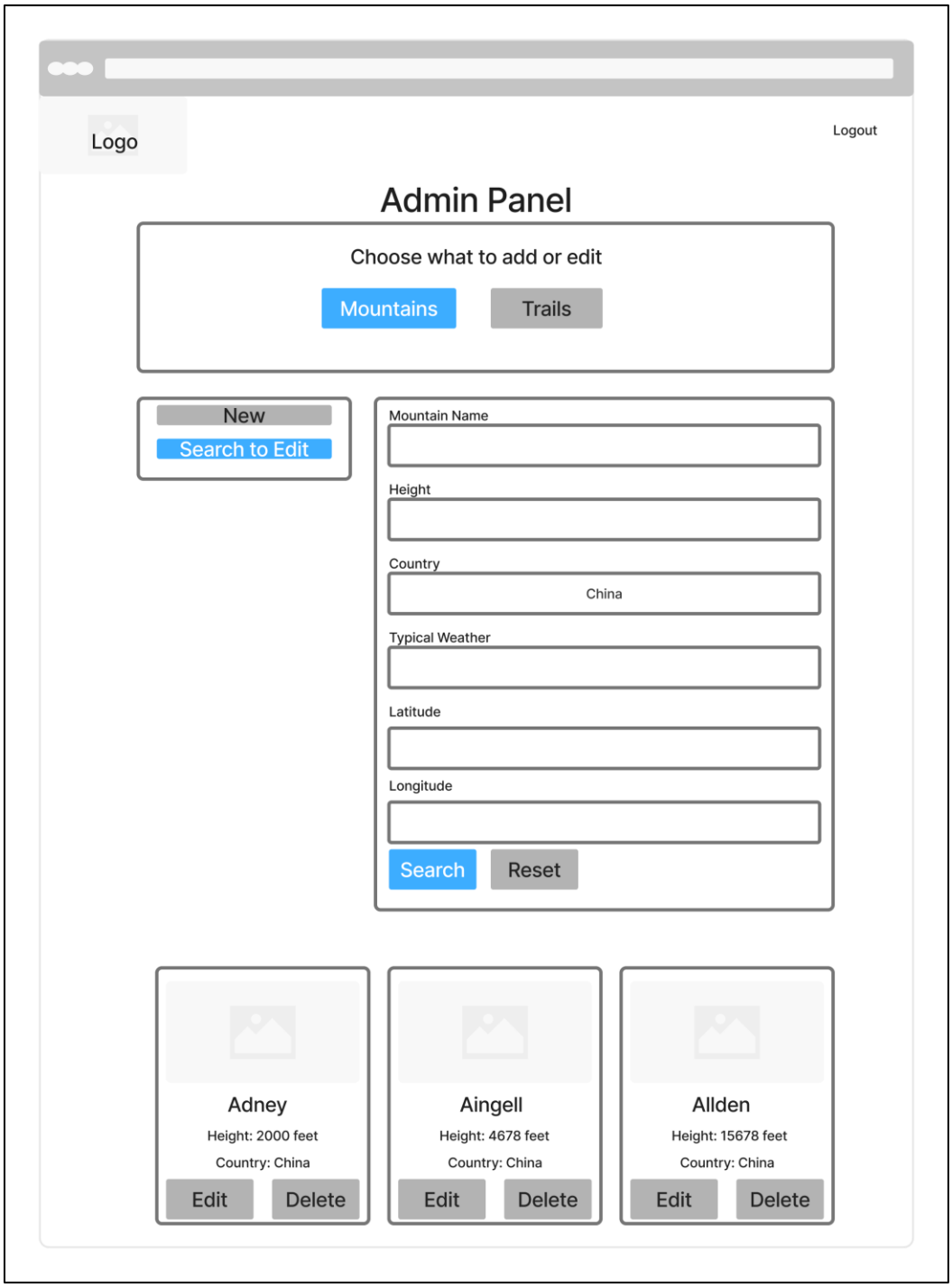
Mockup Page 9 – Admin side – Admin Login (Joseph Sikowitz)



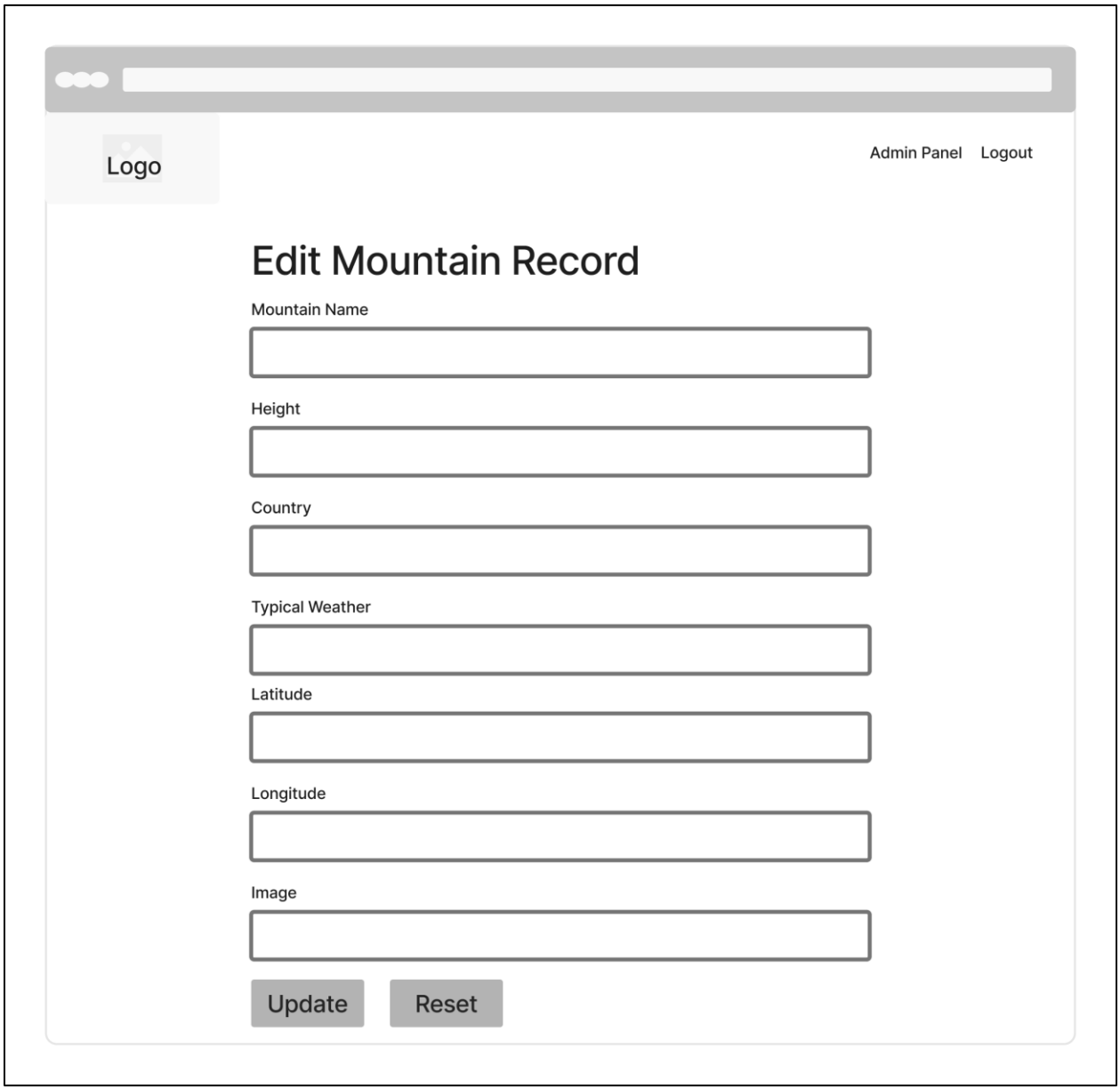
Mockup Page 10 – Admin side – Admin Add Mountain (Joseph Sikowitz)



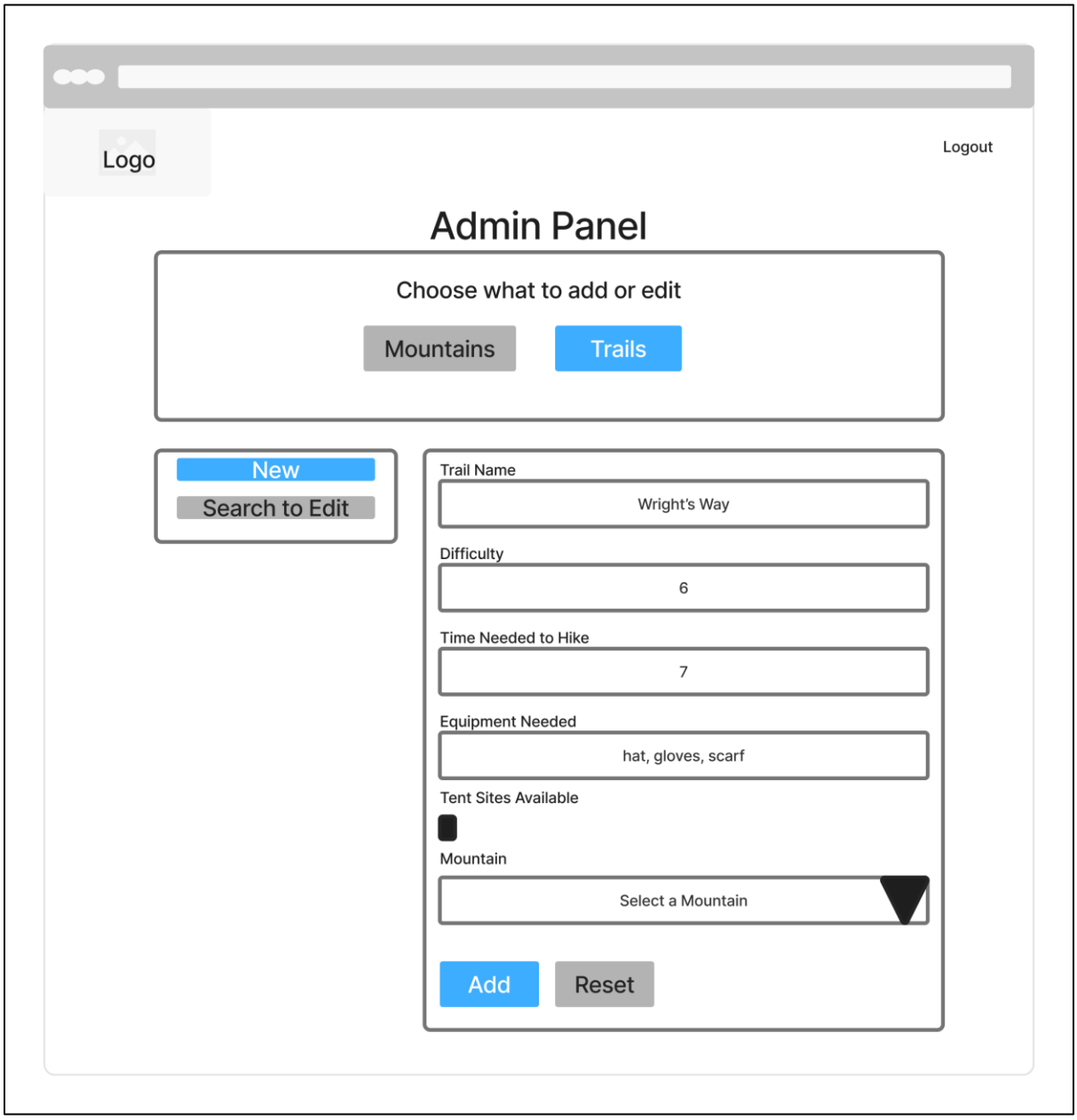
Mockup Page 11 – Admin side – Admin Search to Edit or Delete Mountain (Joseph Sikowitz)



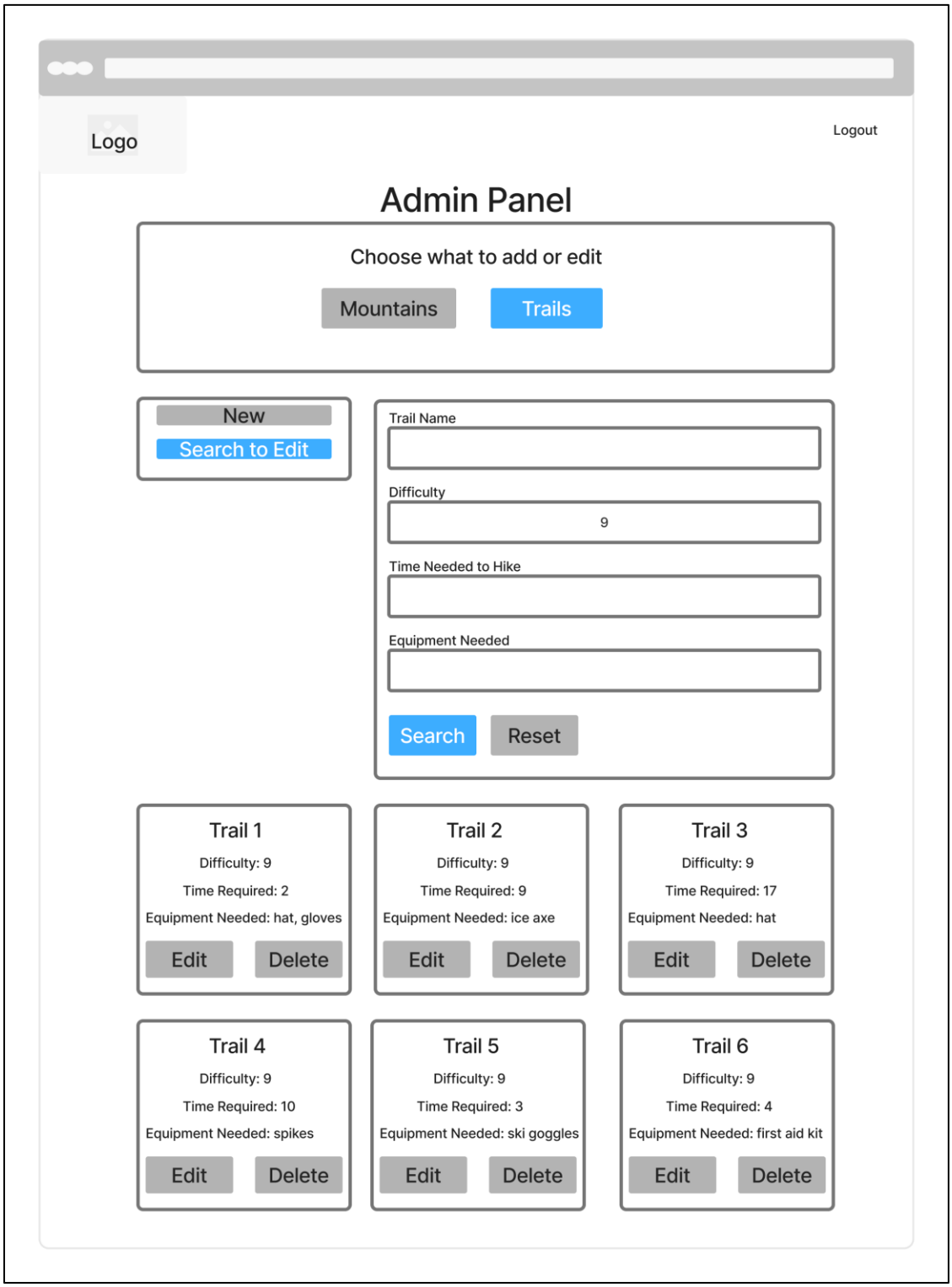
Mockup Page 12 – Admin side – Admin Edit Mountain (Joseph Sikowitz)



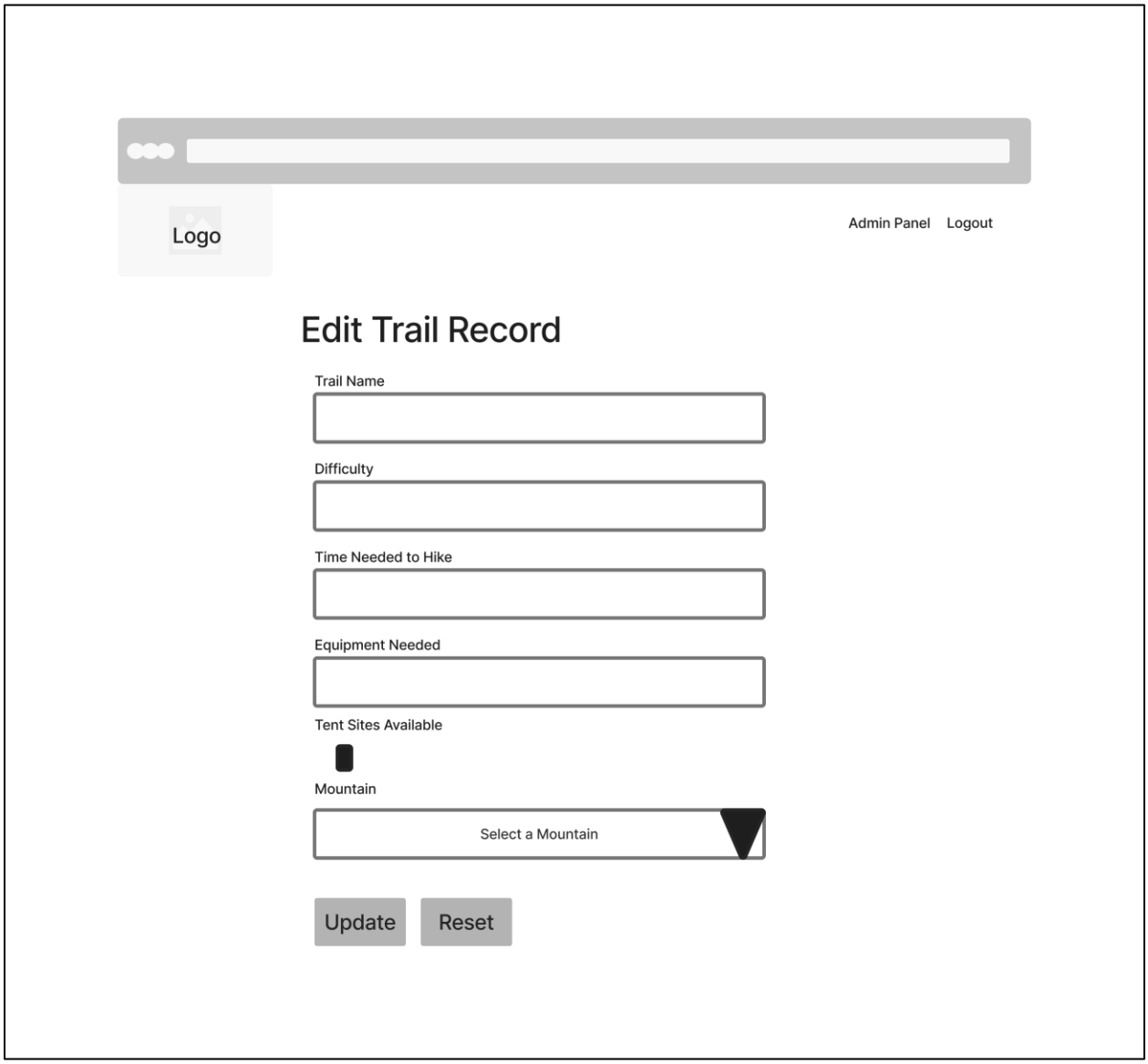
Mockup Page 13 – Admin side – Admin Add Trail (Joseph Sikowitz)



Mockup Page 14 – Admin side – Admin Search to Edit or Delete Trail (Joseph Sikowitz)



Mockup Page 15 – Admin side – Admin Edit Trail (Joseph Sikowitz)



Document Control

Version	Date	Changes Summary	Author
1.0	10/14/2026	First draft	Joseph and Ken